Leasing Solar in Hawai‘i

Individual Research Paper

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The Challenge

Electricity is a major expense for families living in Hawai‘i. According to the U.S. Energy Information Administration (EIA, 2025), Hawai‘i's average electricity price for all customer types reached 38.00 cents per kilowatt-hour in 2024. The average price increased from 27.55 cents in 2020 to 38.00 cents in 2024 (EIA, n.d.), showing how much electricity costs have risen in recent years.

I wanted to research this topic because I have solar at my own home and have seen a difference in my monthly expenses. In December 2023, I paid $362.59 for electricity and $136.59 for my original Sunrun system, totaling $499.18. My current combined monthly cost is about $355.30, including my original Sunrun payment, new solar lease, and remaining electric company charge. My goal is to determine whether leasing solar is a good financial decision over 25 years and whether it could help other homeowners manage the high cost of electricity in Hawai‘i.

Economic Concepts

Three economic concepts that apply to my solar lease are time value of money, substitutes, and inflation. The time value of money means that money today is worth more than the same amount in the future. I make lease payments every month, while the savings from lower electricity costs happen over time, so I need to determine whether those future savings are worth the total cost of the lease. I am focusing on comparing a solar lease with continuing to buy electricity from the electric company.

Substitutes apply because leasing solar is one way to meet my household's electricity needs, while buying electricity from Hawaiian Electric is another option. Inflation is also

important because electricity rates may increase over time, while my lease payment increases by about 2.99% annually. Comparing these increases can help me understand whether leasing solar will continue to save money over the 25-year contract.

Today's Numbers

In December 2023, my monthly electricity costs were $362.59 for my Hawaiian Electric bill and $136.59 for my original Sunrun system. Together, I was paying $499.18 per month. Based on my most recent bills, my costs are now $136.59 for my original Sunrun system, $217.84 for my new solar lease, including tax, and $0.87 for my electric company bill. This brings my current total to $355.30 per month.

The difference between these two amounts is $143.88 per month.

One thing I need to be careful about is that this comparison uses my December 2023 costs and my most recent bill rather than a full year of data. Electricity usage can change depending on the month, household activities, and other factors. These numbers give me a starting point, but I need more information to determine whether the savings will continue over the full 25-year lease.

What Happens Next

To understand how my solar lease could affect my finances over 25 years, I compared four possible annual increases in electricity prices: 0%, 3%, 5%, and 8.4%. The 3% scenario is based on the approximate annual increase in Hawai‘i's average electricity prices from 2010 to 2024. The 8.4% scenario reflects the faster increase from 2020 to 2024. My calculations use an average monthly electricity bill of $337.32, based on my bills from November 2023, December 2023, and January 2024.

Figure 1 shows that my estimated cumulative savings over 25 years are about $8,500 if electricity prices do not increase, $59,000 with 3% annual increases, $110,000 with 5% increases, and $247,000 with 8.4% increases.

The 0% scenario is especially important. Even if electricity prices stay flat, the lease is estimated to save money overall. However, the savings do not keep increasing throughout the contract. The lease saves money for about the first 16 years, and starting in year 17, the lease payment becomes higher than the electric bill it replaces. My cumulative savings peak at approximately $13,500 before falling to about $8,500 by the end of the 25-year period. This shows why I need to consider the timing of payments and savings, not just the monthly cost today.

Based on the model, electricity prices would need to fall by about 0.7% annually for the lease to lose money over the full 25 years. However, these estimates assume that household electricity use and solar performance remain consistent. Changes in my family's energy use, system performance, or future electricity charges could affect the actual savings.

Recommendation

Based on my research, I believe homeowners in Hawai‘i should consider leasing solar, but they should compare the full costs before signing a long-term contract. My model estimates about $8,500 in savings over 25 years if electricity prices stay flat, with higher estimated savings if prices increase. One rule homeowners can use is to compare the lease's annual payment increase with the annual electricity price increase they reasonably expect. My lease payment increases by about 2.99% per year, so the financial benefit is more likely to grow if electricity prices rise faster than that. However, the flat-price scenario shows that the lease payment becomes higher than the electric bill it replaces around year 17, even though the model still

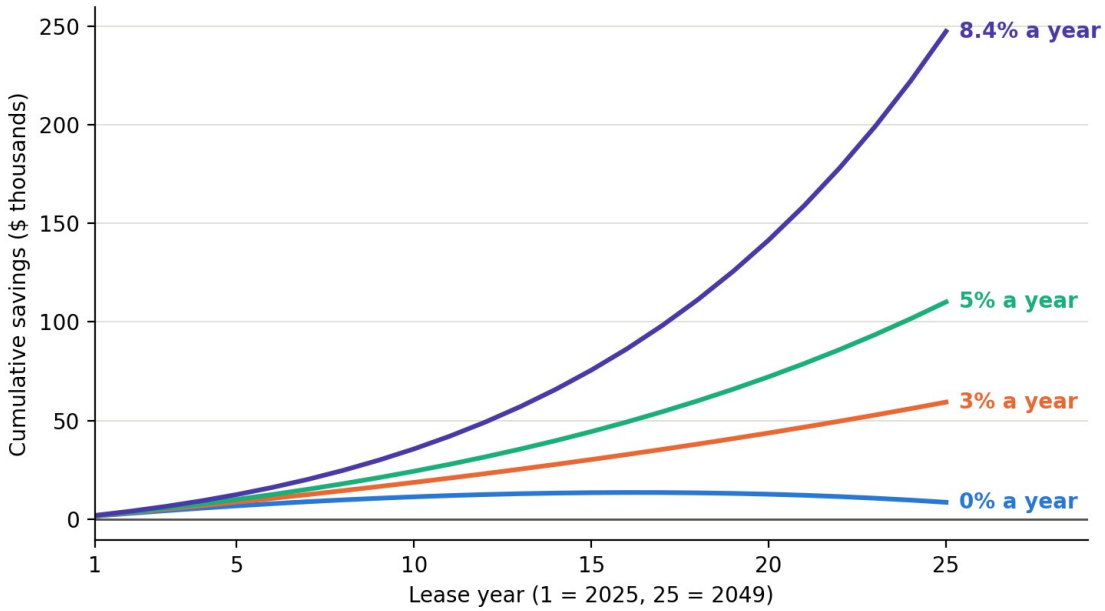
estimates about $8,500 in total savings over 25 years. I would consider leasing if the estimated savings remain positive under conservative assumptions, the payments fit the household budget, and the contract terms are acceptable. Homeowners should also consider how long they plan to stay in the home and compare leasing with other options before deciding.

The Strongest Objection

The strongest argument against leasing solar is that a 25-year contract is a long commitment, and the actual savings may be different from what the chart predicts. My model assumes that household electricity use and solar performance remain consistent and that electricity prices follow the selected scenarios. In reality, my family's energy use could change, the system could produce less electricity, or future costs could be different. The flat-price scenario also shows that the lease payment becomes higher than the electric bill it replaces around year 17, even though the model still estimates about $8,500 in total savings over 25 years. My response is that this does not automatically make leasing a bad decision, but it shows why homeowners should not focus only on their current monthly savings. They should compare the full 25-year costs and test different price assumptions before signing. My experience provides a useful starting point, but each household needs to evaluate its own situation.

Figure 1

Estimated Cumulative Savings From the New Lease Compared With Paying the Electric Company, Under Four Yearly Electricity Price Increases



Note. Assumes an electric bill of $337.32 per month in 2024 (average of the November 2023 to January 2024 bills), a lease of $202.00 per month rising 2.99% per year plus tax, $0.87 per month still owed to the electric company, and the same electricity use every year.

References

U.S. Energy Information Administration. (n.d.). Average price by state by provider (EIA-861) [Data set]. https://www.eia.gov/electricity/data/state/avgprice_annual.xlsx

U.S. Energy Information Administration. (2025, November 10). Hawaii: State profile and energy estimates. https://www.eia.gov/electricity/state/hawaii/